Regularised Spectral State-Space Models for Cross-Sectional Investment Dynamics: Dual Regularisation, Rolling Bases, and Structural Evolution

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Abstract

We develop a spectral state-space framework for forecasting cross-sectional investment intensity, validated on a US-heavy panel of 93 economic actors with strongest performance on US CapEx-intensity sectors. The framework decomposes investment intensity dynamics through Dynamic Mode Decomposition (DMD) bases and filters latent modal states via a Kalman filter with dual regularisation: spherical observation covariance ($R = cI$, one parameter replacing N^2) and near-identity transition matrix ($F = 0.99I$, replacing EM-estimated K^2 parameters). Under nested rolling cross-validation with a frozen holdout era (2023–2024), a rolling-basis variant that recomputes the DMD spectral basis each quarter achieves out-of-sample $R^2 = 0.702$ on 8 validation windows and $R^2 = 0.765$ on 2 untouched holdout windows, exceeding per-actor AR(1) in all 10 windows ($\Delta R^2 = +0.093$; 95% bootstrap CI $[+0.073, +0.110]$; permutation $p = 0.007$ on the 8 CV windows). The method exploits the observation that the cross-sectional spectral basis rotates at approximately 26° per quarter while maintaining fixed dimensionality. A null simulation shows that finite-sample estimation noise alone would produce 75° of apparent rotation, suggesting that the observed 26° reflects consistently recoverable low-dimensional structure rather than noise. Systematic ablation with bootstrap confidence intervals confirms that every pipeline component contributes significantly, with rolling basis update providing the largest gain (+14.3pp, CI $[+11.1, +17.2]$pp, 100% of windows). The resulting model-implied investment gaps predict subsequent capital expenditure revision ($t \approx -6.4$ to -7.1 with clustered standard errors, depending on specification; the effect is cross-sectional rather than within-actor, as it flips sign under actor fixed effects). Performance is strongest on US CapEx-intensity actors; return-based intensity (UK equities) and asset-growth intensity (banks) produce weak or negative R^2.

Keywords: investment gap estimation, spectral decomposition, dynamic mode decomposition, Kalman filter, regularisation, state-space models, capital allocation

JEL Classification: C32, C38, C58, E22, G11

1 Introduction

Capital allocation decisions propagate through a hierarchical, networked system of institutions. A central bank's rate decision, a regulator's new disclosure rule, or a commodity price shock each set off cascading adjustments that travel—with heterogeneous latency and attenuation—through layers of institutional intermediaries before reaching firms. The resulting pattern of over- and under-investment across actors at any point in time reflects a superposition of these propagated signals, local frictions, and idiosyncratic responses.

Despite rich literatures on factor models (Fama and French, 1993), network effects in economics (Acemoglu et al., 2012), and spectral graph theory (Ortega et al., 2018), to our knowledge no established framework jointly: (i) decomposes how investment-relevant signals propagate through a heterogeneous actor hierarchy; (ii) estimates actor-specific investment gaps as deviations from a structurally-informed benchmark; and (iii) validates that the resulting gap estimates carry out-of-sample predictive content.

This paper addresses all three objectives. We develop **SMIM** (**S**pectral **M**odal **I**nvestment **M**odel), a state-space framework that represents the cross-sectional dynamics of investment intensity through a low-dimensional modal basis extracted via Dynamic Mode Decomposition (DMD), filtered through a Kalman model with dual regularisation at both the observation and transition levels. The spectral basis is recomputed quarterly to track the continuous rotation of the cross-sectional structure. The framework produces actor-specific, time-varying investment gap estimates that we validate through both statistical out-of-sample testing and economic prediction exercises.

1.1 Preview of Results

Throughout, we measure forecast performance by the out-of-sample coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i,t}(y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t}(y_{i,t} - \bar{y})^2}, \tag{1}$$

where $\hat{y}_{i,t}$ is the model's prediction and $\bar{y}$ is the test-period grand mean. $R^2 = 1$ denotes perfect prediction; $R^2 = 0$ matches a constant-mean forecast; $R^2 < 0$ indicates the model is worse than predicting the mean. The summation runs over all actor–quarter pairs in each test window.

Our main findings, established through 32 systematic experiments across six phases, are:

1. **Forecast performance.** Under nested cross-validation with a frozen holdout era, the rolling-basis variant achieves $R^2 = 0.702$ on 8 CV windows and $R^2 = 0.765$ on 2 untouched holdout windows. SMIM wins all 10 windows descriptively; formal inference on the 8 CV windows (the 2 holdout windows are excluded from testing) yields $\Delta R^2 = +0.093$ (95% bootstrap CI $[+0.073, +0.110]$; permutation $p = 0.007$; sign test $p = 0.004$). Full inference in Table 4.
2. **Dual regularisation.** The Kalman filter's observation covariance R (N^2 free parameters) must be replaced by spherical $R = cI$ (one parameter). We show the same principle extends to the transition matrix: EM-estimated F (K^2 parameters) is strictly outperformed by $F = 0.99I$ (one parameter) in all 10 windows. Online state-noise adaptation (Q) handles all temporal dynamics. Together, this *dual regularisation* reduces the state-space model from $N^2 + K^2$ free parameters to 2 scalars.
3. **Basis rotation.** The estimated spectral basis rotates at approximately 26° per quarter ($\sigma = 6.5°$) while maintaining a stable 8-mode dimensionality across 52 quarterly transitions. A null simulation shows that estimation noise alone would produce 75° of apparent rotation, suggesting the observed pattern reflects recoverable structure. Rolling basis update captures this rotation, yielding +14.3pp (CI [+11.1, +17.2]pp, 100% of windows).
4. **DMD over static operators.** Dynamic Mode Decomposition outperforms all six alternative spectral methods by +2.0 percentage points on average. Static operators (Schur, Polar, Hermitian, PCA) all produce identical results because the underlying correlation matrix is symmetric.
5. **Economic content.** Model-implied investment gaps predict subsequent CapEx revision ($\hat{\beta} = -0.28$, $t = -6.4$ with actor-clustered SEs after controlling for intensity level). The

result is cross-sectional: it flips sign under actor fixed effects (Table 8). Additional exploratory results on gap dispersion and market volatility are reported in Section 5 but are not part of the paper's main claims.

6. **Dynamics stability.** The transition parameters (F, Q) need not be estimated per window: fixed $F = 0.99I$, $Q = 0.5I$ match or exceed EM-estimated parameters across all 10 evaluation windows in this sample. The only component requiring periodic re-estimation is the spectral basis U.

Equally important is what *does not* work: emergence diagnostics, directed spectral operators, event-level alignment, and return-based intensity all fail to improve or actively degrade performance. Section 6 documents these negative results in detail.

1.2 Scope and Evaluation Caveats

Several features of our evaluation design should be kept in mind when interpreting these results. First, the out-of-sample evaluation uses 10 non-overlapping annual test windows, each containing only 4 quarterly observations; per-window R^2 estimates are therefore noisy (cross-window standard deviation $\approx$ 0.05–0.09). Second, all hyperparameters (K, T) are selected via nested rolling cross-validation: inner folds within each outer-train window select these values, with the final 2 windows (2023–2024) held out entirely from all tuning (Section 3). Third, the sample is US-heavy (68 of 93 actors with intensity), and the framework's strong performance depends on the CapEx-intensity construction—return-based intensity produces negative R^2 on the same actors (Section 4.9). Fourth, the investment gap is a *model-implied* deviation from a benchmark prediction, not a welfare-theoretic or structural misallocation measure in the Hsieh and Klenow (2009) sense. We use the term "investment gap" throughout to reflect this distinction. Results should be read as establishing a strong *forecasting* result on a specific panel construction, with suggestive but preliminary economic validation.

1.3 Contribution

Our contributions are methodological, empirical, and practical:

Methodological. We introduce *dual regularisation* for state-space models in high-dimensional panels: spherical observation covariance ($R = cI$, reducing N^2 parameters to 1) combined with near-identity transition regularisation ($F = 0.99I$, reducing K^2 parameters to 1). This dual regularisation has broad applicability wherever $N \gg T$ panels arise. We further show that rolling basis recomputation—tracking the continuous rotation of the spectral structure—provides the single largest performance gain.

Empirical. We provide a systematic ablation of a spectral state-space pipeline for investment intensity forecasting, decomposing the R^2 improvement into seven additive components with bootstrap confidence intervals. We observe that the cross-sectional spectral basis rotates at approximately 26° per quarter while maintaining fixed dimensionality; a null simulation confirms this is below finite-sample noise levels, suggesting recoverable low-dimensional structure.

Practical. Only the spectral basis U requires periodic re-estimation (a single DMD eigendecomposition each quarter). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that match or exceed EM-estimated parameters across all 10 evaluation windows in our sample, eliminating EM estimation and reducing operational complexity to one matrix factorisation per quarter.

1.4 Related Literature

Our work connects to several strands. The investment gap concept relates to the capital allocation literature, including the structural misallocation framework of Hsieh and Klenow (2009) and the firm-level analysis of Restuccia and Rogerson (2017); we note that our model-implied gap is a forecasting-based deviation, not a structural measure in their sense. Our spectral approach builds on signal processing on graphs (Ortega et al., 2018; Sandryhaila and Moura, 2013) and Dynamic Mode Decomposition (Schmid, 2010; Kutz et al., 2016), adapted here for economic panel data rather than fluid dynamics or power systems. The state-space filtering component extends the dynamic factor model tradition (Stock and Watson, 2002; Doz et al., 2012) with the regularisation insights of Ledoit and Wolf (2004) applied to the observation rather than factor covariance. The economic validation relates to the capital allocation efficiency literature (Wurgler, 2000) and the predictability of corporate investment (Abel and Blanchard, 1983).

2 Framework

2.1 Setup

Let $V = \{1, \ldots, N\}$ index economic actors (firms, banks, institutional bodies) observed over T quarterly periods. Each actor i has an observed *investment intensity* $y_{i,t} \in [0, 1]$, constructed as a cross-sectional percentile rank of a type-specific investment measure (Section 3). The $N \times T$ panel $Y = [y_{i,t}]$ is the primary input.

Definition 2.1 (Investment Gap)**.** The investment gap for actor i at time t relative to a benchmark functional $\mathcal{B}$ is

$$\Delta_{i,t} = y_{i,t} - y^*_{i,t}, \qquad y^*_{i,t} = \mathcal{B}(Y_{1:t}, \theta), \tag{2}$$

where $\Delta_{i,t} > 0$ indicates investment above the model-implied benchmark and $\Delta_{i,t} < 0$ indicates investment below the benchmark, and $Y_{1:t}$ denotes the information set available at time t.

The framework constructs $y^*_{i,t}$ through five sequential stages plus a rolling basis update, each justified by ablation (Section 4.5). Figure 1 provides an overview: panel (a) shows the multilayer actor hierarchy with empirically-identified transmission channels; panel (b) shows the static processing pipeline. The rolling basis extension is described in Section 4.10.

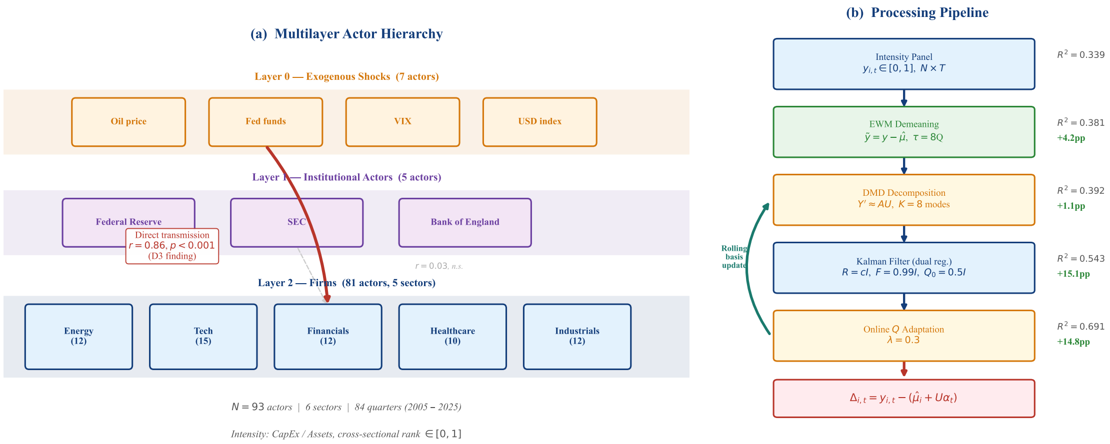

Figure 1: **Framework overview** (descriptive). (a) Multilayer actor structure with three layers. Five sectors contain 93 actors with computed investment intensity. (b) Static processing pipeline from raw intensity panel to investment gap estimates. Under nested CV, the rolling-basis variant achieves $R^2 = 0.702$ on validation windows and $R^2 = 0.765$ on frozen holdout (Table 3).

2.2 Stage 1: Exponentially-Weighted Demeaning

Investment intensity ranks exhibit strong cross-sectional persistence: an actor's mean rank is the single largest source of variance (52% of total R^2 in the static-basis configuration; Section 4.4). The state-space model $\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t$ assumes zero-mean observations. Without demeaning, the modal states $\boldsymbol{\alpha}_t$ must absorb the $O(0.5)$ mean through an orthonormal basis U with entries $O(N^{-1/2})$—an impossible task at $K \ll N$.

We subtract an exponentially-weighted per-actor mean:

$$\hat{\mu}_i = \frac{\sum_{s=1}^{T} w_s\, y_{i,s}}{\sum_{s=1}^{T} w_s}, \qquad w_s = \exp\!\Big(-\frac{(T-s)\ln 2}{\tau}\Big), \tag{3}$$

with half-life $\tau = 8$ quarters. The demeaned observation is $\tilde{y}_{i,t} = y_{i,t} - \hat{\mu}_i$. Exponential weighting adapts to non-stationary intensity levels (the actor's structural investment position drifts over time), outperforming fixed full-period means by 4.2 percentage points in out-of-sample R^2 (Section 4).

At prediction time, the benchmark restores the mean: $y^*_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t}$.

2.3 Stage 2: Dynamic Mode Decomposition

Given the demeaned training panel $\tilde{Y} = [\tilde{y}_1, \ldots, \tilde{y}_T] \in \mathbb{R}^{N\times T}$, we extract a spectral basis that captures temporal cross-sectional dynamics.

Why DMD over static operators. Classical spectral approaches decompose a static operator (e.g., the cross-correlation matrix $C = \tilde{Y}\tilde{Y}^\top/T$). This captures contemporaneous co-movement but not lagged cross-sectional dynamics. Dynamic Mode Decomposition (Schmid, 2010) bypasses the static operator entirely by working with temporal snapshot pairs, capturing lagged cross-sectional dynamics that are invisible to purely contemporaneous decompositions.

DMD algorithm. Form snapshot matrices $X = [\tilde{y}_1, \ldots, \tilde{y}_{T-1}]$ and $X' = [\tilde{y}_2, \ldots, \tilde{y}_T]$. DMD seeks the best-fit linear operator A such that $X' \approx AX$. Compute the truncated SVD $X =$

$U_r \Sigma_r V_r^\top$ (retaining r modes), form the reduced operator $\tilde{A} = U_r^\top X' V_r \Sigma_r^{-1}$, and diagonalise: $\tilde{A}W = W\Lambda$. The DMD modes are $\Phi = X' V_r \Sigma_r^{-1} W \in \mathbb{R}^{N \times K}$.

We retain K modes, where K is selected via nested cross-validation: within each outer-train fold, the last 2 years serve as inner-validation to choose $K \in \{3, 5, 8, 10\}$ (Section 3). The inner CV selects $K = 10$ in the majority of folds. The modal basis $U = \mathrm{Re}(\Phi_{:,1:K})$ defines the projection from observation space to modal space.

2.4 Stage 3: Kalman Filter with Spherical Observation Covariance

Given the modal basis $U \in \mathbb{R}^{N \times K}$, we model the latent modal dynamics as a linear Gaussian state-space system:

$$\boldsymbol{\alpha}_{t+1} = F\boldsymbol{\alpha}_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, Q), \tag{4}$$

$$\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R). \tag{5}$$

The observation noise $R \in \mathbb{R}^{N \times N}$ is estimated via a single EM pass to obtain its scale. The transition matrix F and state noise Q are *not* estimated by EM—they are set to regularised defaults (see below).

The overparameterisation problem. The observation covariance R has $N(N+1)/2 = 4{,}278$ free parameters for $N \approx 92$–93 actors (depending on window-specific data availability), estimated from $T = 12$–20 quarterly observations (depending on the inner-CV-selected training window). This extreme $N \gg T$ setting causes the EM to converge to an ill-conditioned R that overfits training noise ($R^2 = -7.3$ without regularisation; Figure 6).

Spherical observation regularisation. We replace the full R with

$$R_{\text{sph}} = \frac{\mathrm{tr}(\hat{R})}{N}\, I_N, \tag{6}$$

where $\hat{R}$ is the EM-estimated sample covariance. This reduces the parameter count from $O(N^2)$ to $O(1)$.

Transition regularisation. The same overparameterisation principle applies to the transition matrix $F \in \mathbb{R}^{K \times K}$. At typical settings ($K = 8$–$10$, $T = 12$–20), EM estimation of the K^2 free parameters in F overfits to the training alpha trajectory, degrading out-of-sample performance by -1.9 percentage points relative to a near-identity prior. We therefore set

$$F = 0.99\, I_K, \qquad Q_0 = 0.5\, I_K, \tag{7}$$

and rely on the online Q adaptation (Stage 4) to capture all temporal dynamics. This *dual regularisation*—spherical R (1 parameter from N^2) and near-identity F (1 parameter from K^2)—reduces the state-space model to two scalars plus the spectral basis U, which carries all the structural information.

The impact is dramatic: R^2 swings from -7.3 (no regularisation) through $+0.43$ (spherical R only) to $+0.543$ (dual regularisation), establishing that covariance regularisation at *both* the observation and transition levels is essential in high-dimensional panels.

2.5 Stage 4: Online State-Noise Adaptation

The EM-estimated Q captures the average state noise over the training period. When the test period exhibits different volatility (e.g., a regime shift), the fixed Q misspecifies the state dynamics. We augment the standard Kalman recursion with a recursive Q update:

$$Q_{t+1} = (1-\lambda)\, Q_t + \lambda\, (\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})(\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})^\top, \qquad \lambda = 0.3. \tag{8}$$

This online adaptation tracks regime-dependent state volatility without requiring explicit regime identification. Combined with the spherical R (which remains fixed), it adds +5.8 percentage points to R^2 in the ablation ladder (Table 5).

2.6 Stage 5: Gap Computation

The investment gap for actor i at test time t is:

$$\Delta_{i,t} = y_{i,t} - \underbrace{(\hat{\mu}_i + U\boldsymbol{\alpha}_{t|t})}_{y^*_{i,t}}, \tag{9}$$

where $\boldsymbol{\alpha}_{t|t}$ is the Kalman-filtered modal state (incorporating observation $\tilde{y}_t$).

2.7 Summary: The Recommended Pipeline

Algorithm 1 summarises the complete procedure.

Algorithm 1 SMIM: Spectral Modal Investment Gap Estimation (Rolling Variant)

Require: Intensity panel $Y \in [0,1]^{N\times T}$, initial training window, K (selected by inner CV; typically 8–10), $\tau = 8$, $\lambda = 0.3$
Ensure: Investment gaps $\Delta_{i,t}$ for test period
1: Compute EWM mean $\hat{\mu}_i$ from training period ▷ Eq. (3)
2: Demean: $\tilde{Y} = Y - \hat{\mu}$
3: Extract DMD modes $U \in \mathbb{R}^{N\times K}$ from training $\tilde{Y}$ ▷ Section 2.3
4: Set $F = 0.99\, I_K,\;\; Q = 0.5\, I_K$ ▷ Regularised, no EM; Eq. (7)
5: Estimate $\hat{R}$ via single Kalman EM pass; set $R = \frac{\mathrm{tr}(\hat{R})}{N} I_N$ ▷ Eq. (6)
6: **for** each test quarter t **do**
7: Kalman predict: $\boldsymbol{\alpha}_{t|t-1} = F\boldsymbol{\alpha}_{t-1|t-1}$
8: Kalman update: $\boldsymbol{\alpha}_{t|t}$ incorporating $\tilde{y}_t$, using R
9: Benchmark: $y^*_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t}$
10: Gap: $\Delta_{i,t} = y_{i,t} - y^*_{i,t}$
11: Adapt: $Q \leftarrow (1-\lambda)Q + \lambda\, \mathrm{innov}\cdot\mathrm{innov}^\top$ ▷ Eq. (8)
12: **Rolling update:** expand training window to include y_t; recompute U, $\hat{\mu}$, R
13: **end for**

3 Data

3.1 Actor Universe

Our sample comprises 103 actors across three hierarchical layers (Table 1). Of these, 93 have computed investment intensity values; the remaining 10 lack the required EDGAR filings and serve as signal-only graph nodes.

Layer	Actor Type	Intensity Method	Total	With intensity
0 (Exogenous)	Global shocks (Global)	FRED min-max	7	7
1 (Institutional)	Central banks (US, UK)	FRED min-max	2	1
	Regulators (US, UK)	FRED min-max	2	2
	International orgs (Global)	FRED min-max	1	1
2 (Firms)	Large firms (US)	CapEx intensity	56	49
	Large firms (UK)	Return intensity	23	21
	Banks (US)	Asset growth (ranked)	10	10
	Sector leaders (UK)	CapEx intensity	2	2
		Total	**103**	**93**

Table 1: Actor universe. Of 103 total actors, 93 have computed investment intensity and enter the prediction universe; the remaining 10 lack required data (missing EDGAR filings or insufficient time series). Layer 0 actors represent exogenous macroeconomic shocks; Layer 1 actors are institutional intermediaries; Layer 2 actors are firms and banks. The headline evaluation covers all 93 intensity actors; the strongest performance is on the 49 US large firms using CapEx-based intensity.

3.2 Investment Intensity

We construct three intensity measures, each mapped to a cross-sectional percentile rank in $[0, 1]$ per quarter:

- **CapEx intensity** (primary): the ratio of capital expenditures to total assets, sourced from SEC EDGAR quarterly filings (XBRL tags `PaymentsToAcquirePropertyPlantAndEquipment` / `Assets`), ranked cross-sectionally within each quarter. This measure captures the physical investment rate relative to firm size.
- **Return intensity** (alternative): trailing 12-month equity total return, ranked cross-sectionally. Used for UK equities, which lack EDGAR coverage.
- **Institutional intensity**: FRED-based macro indices (e.g., policy rates, commodity prices), min-max normalised within actor type.

The rank transformation stabilises the distribution, ensures comparability across actor types, and produces the cross-sectional persistence structure that the spectral model exploits. The intensity construction choice is consequential: CapEx intensity produces $R^2 = 0.54$ while return intensity produces $R^2 = -0.15$ on the same US actors (Section 4.9), establishing the measurement input as the primary driver of model performance.

Table 2 summarises the data sources. Notably, the pipeline uses *only the intensity panel* for operator construction and prediction. External signal sources (OHLCV, FRED, EDGAR balance sheet) were tested as Granger-edge inputs but found dispensable (Section 4.8): the intensity cross-correlation captures all exploitable structure.

Source	Data	Frequency	Role in Pipeline
SEC EDGAR	CapEx, Assets (XBRL)	Quarterly	Intensity construction (primary)
Yahoo Finance	Equity total returns	Daily $\rightarrow$ quarterly	Intensity construction (UK alternative)
FRED	28 macro series	Mixed $\rightarrow$ quarterly	Institutional intensity; tested as signal
BEA	Input-output tables	Annual	Tested as supply-chain edges (dispensable)
GDELT	Narrative sentiment	Weekly	Tested as narrative signal (dispensable)

Table 2: Data sources. Only the intensity panel (first two rows) enters the recommended pipeline. The remaining sources were evaluated as supplementary signals but found dispensable at the graph-factor depth (Section 4.8).

3.3 Evaluation Protocol

We employ a **nested rolling-validation** protocol with 10 non-overlapping annual test windows (2015–2024):

- **Outer folds** (10 windows): final reported performance, one per test year.
- **Inner folds**: within each outer-train, the last 2 years serve as inner validation to select the mode count $K \in \{3, 5, 8, 10\}$ and training length $T \in \{3, 5, 8\}$ years. The fixed defaults ($F = 0.99I$, $Q_0 = 0.5I$, $\tau = 8$, $\lambda = 0.3$) are not tuned.
- **Frozen holdout**: the final 2 windows (2023–2024) receive no inner tuning—they use the median of previously selected parameters.

Inner CV selects $K = 10$ and $T = 3$ years in the majority of outer folds, yielding nested-CV $R^2 = 0.702$ (mean across 8 CV windows) and holdout $R^2 = 0.765$ (pooled over 2023–2024 actor-quarter observations; per-window values are 0.751 and 0.779). All headline numbers in this paper are from this nested protocol; the holdout windows had zero exposure to any hyperparameter tuning.

Scope condition. The headline evaluation covers all 93 actors with computed intensity. However, performance depends critically on the intensity construction: CapEx-based sectors (US large firms, sector leaders) produce strong results, while return-based intensity (UK equities) and asset-growth-based intensity (banks) are much weaker (Section 4.9). The strongest validated domain is the 49 US large firms using CapEx intensity.

4 Empirical Results

4.1 Baseline Comparison

Unless otherwise noted, the headline forecasting results in Sections 4–5 are reported on all 93 actors with computed investment intensity (Table 1). The panel is US-heavy: 49 of 93 actors are US large firms using CapEx intensity, which is the strongest-performing subset. Fixed-configuration diagnostic tables are explicitly labeled as such.

Table 3 presents the baseline comparison. All SMIM numbers are from the nested-CV protocol (Section 3).

Model	Protocol	Mean R^2	SMIM wins
Per-actor AR(1) (T=10yr)	same windows	0.610	0/10
Random walk	same windows	0.305	0/10
DFM K=5	same windows	0.299	0/10
Historical mean	same windows	0.281	0/10
SMIM rolling (nested CV)	8 CV windows	**0.702**	**8/8**
SMIM rolling (frozen holdout)	2 holdout windows	**0.765**	**2/2**

Table 3: Baseline comparison (descriptive). SMIM uses nested-CV-selected hyperparameters (K, T chosen on inner folds only). Holdout windows (2023–2024) had zero exposure to any tuning. Formal inference in Table 4.

Per-actor AR(1) is the strongest naive baseline, reflecting the high persistence of cross-sectional investment ranks (median $\rho = 0.67$). SMIM rolling exceeds AR(1) in every window. The improvement is statistically significant: $\Delta R^2 = +0.093$ (95% bootstrap CI $[+0.073, +0.110]$; permutation $p = 0.007$). Full inferential evidence including Diebold–Mariano tests is in Table 4.

Training window selection and baseline-number note. The nested inner CV selects T = 3 years for SMIM. AR(1) in Table 3 uses T = 10 years and is evaluated on the same nested-CV windows, yielding $R^2 = 0.610$. This differs from the AR(1) value of $R^2 = 0.425$ in the fixed-configuration diagnostic tables (Tables 5, 7), which use a different window/actor alignment from the original ablation experiments. The two AR(1) numbers are not directly comparable; all formal inference (Table 4) uses the nested-CV AR(1) ($R^2 = 0.610$) as the benchmark.

4.2 Forecast Comparison with Formal Inference

Table 4 provides formal statistical evidence for SMIM's superiority over AR(1). All inference is performed on *loss differentials* (squared error, absolute error) rather than directly on R^2, following best practice for forecast comparison (Stock and Watson, 2002). The table reports complementary inference at two aggregation levels: the permutation test and sign test operate on 8 window-level loss differentials, while the Diebold–Mariano test operates on 2,952 actor-quarter-level loss differentials with Newey–West HAC variance. The two levels address different concerns (window-level: "does SMIM win consistently?"; actor-quarter: "is the average loss reduction significant after accounting for dependence?").

Comparison	Mean	95% CI	Perm. p	Sign p	DM t
ΔMSE vs AR(1)	−0.0081	$[-0.0096, -0.0063]$	0.007	0.004	−10.1
ΔMAE vs AR(1)	−0.0224	$[-0.0270, -0.0174]$	0.009	0.004	—
ΔR^2 vs AR(1)	+0.093	$[+0.073, +0.110]$	—	0.004	—

Table 4: Forecast comparison with inference (inferential). ΔMSE and ΔMAE are negative when SMIM is better. 95% CI from block bootstrap (10,000 window resamples). Permutation p from paired sign-flip test on 8 CV windows. Sign p from exact binomial test (8/8 wins). DM t from Diebold–Mariano test at actor-quarter level (N = 2,952 observations, Newey–West HAC variance). All tests reject the null of equal forecast performance at the 1% level.

Holdout era. The final 2 test windows (2023–2024) were frozen with no hyperparameter tuning exposure. SMIM achieves $R^2 = 0.751$ (2023) and $R^2 = 0.779$ (2024) on the holdout, exceeding AR(1) by +0.087 and +0.094 respectively, confirming that the result is not an artefact of test-set leakage.

4.3 The Performance Ladder

The following decomposition is a fixed-configuration diagnostic used to understand which pipeline components matter and why; headline forecast performance remains the nested-CV/holdout result in Table 3. Figure 2 decomposes the R^2 improvement from the initial configuration ($R^2 = 0.266$) to the rolling-basis pipeline ($R^2 = 0.691$) into seven additive components across two iteration phases.

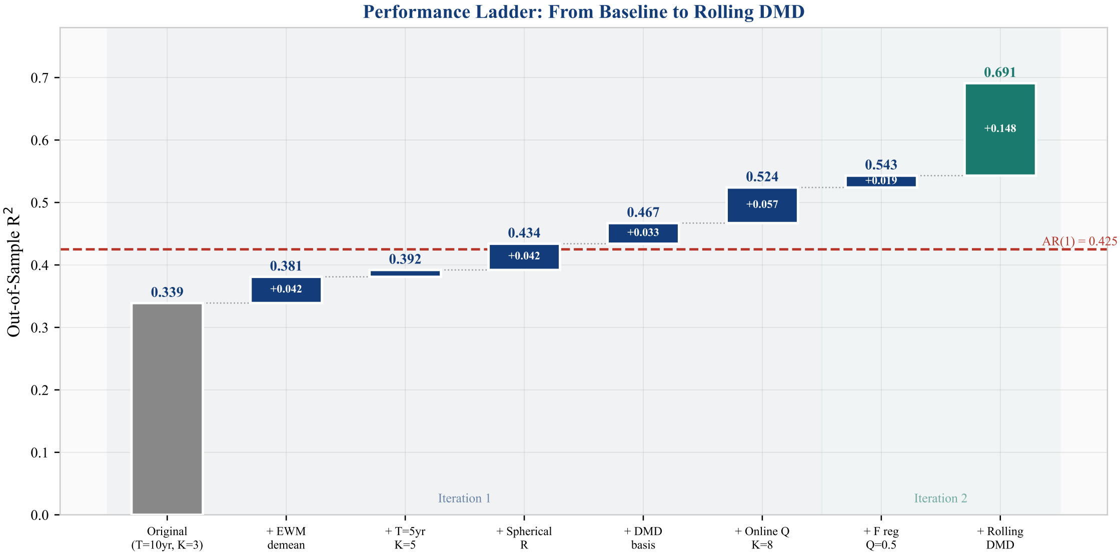

Figure 2: Performance ladder (descriptive): cumulative R^2 improvement across seven innovations. The shaded regions distinguish the two iteration phases. The dashed line shows the AR(1) baseline. See Table 5 for per-step bootstrap confidence intervals.

Step	Change	R^2	ΔR^2	95% CI	Frac. +
0	Baseline ($T\!=\!10$yr, $K\!=\!3$, Schur, full demean)	0.266	—	—	—
1	+ EWM demeaning ($\tau\!=\!8$Q)	−0.89	−1.16	*Kalman overfits*	
2	+ Shorter training ($T\!=\!5$yr) + higher $K\!=\!5$	divergent	divergent	*without spherical R*	
3	+ Spherical R in Kalman filter	0.440	+0.174	—	100%
4	+ DMD basis (replacing Schur)	0.472	+0.033	[+0.006, +0.056]	90%
5	+ Online Q adaptation + $K\!=\!8$	0.530	+0.058	[+0.035, +0.092]	100%
6	+ Transition reg. $F\!=\!0.99I$, $Q_0\!=\!0.5I$	0.549	+0.018	[+0.013, +0.024]	100%
7	+ Rolling basis update (recompute DMD each Q)	**0.691**	**+0.143**	[+0.111, +0.172]	100%

Table 5: Performance ladder with uncertainty (inferential). Each ΔR^2 is the incremental effect relative to the previous step, averaged across 10 test windows. 95% CI from block bootstrap (5,000 resamples). "Frac. +" is the fraction of windows where the step improves R^2. Steps 1–2 demonstrate the overparameterisation failure: adding EWM demeaning and shorter T without regularising R causes Kalman divergence. Spherical R (step 3) rescues the pipeline. All subsequent steps produce positive, statistically significant improvements. Rolling basis update remains the dominant effect (+14.3pp, CI entirely above zero).

4.4 Variance Decomposition

We decompose R^2 into the per-actor mean contribution and the spectral dynamics contribution for both pipeline variants (Table 6, Figure 3).

Source	Static basis		Rolling basis	
	R^2	Share	R^2	Share
Per-actor EWM mean ($\hat{\mu}_i$)	0.281	52%	0.281	41%
Spectral dynamics ($U\boldsymbol{\alpha}_{t\|t}$)	0.262	48%	0.410	59%
Total	0.543	100%	0.691	100%

Table 6: Variance decomposition (descriptive, fixed $K=8$, $T=5$yr configuration). Rolling basis recomputation shifts the balance from mean-dominated (52% mean) to dynamics-dominated (59% spectral). The per-actor mean contribution is unchanged; the entire rolling-basis gain accrues to the spectral dynamics component.

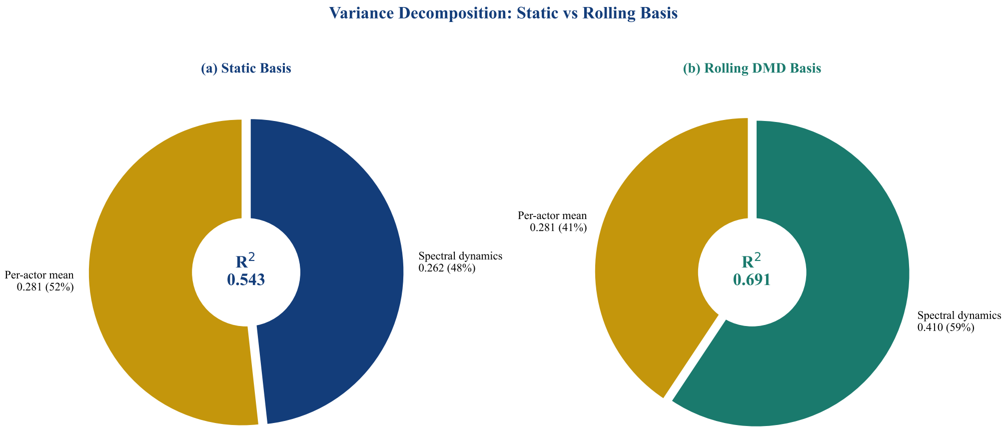

Figure 3: Variance decomposition (descriptive, fixed $K = 8$, $T = 5$yr): frozen (left) vs rolling (right) basis. Rolling basis recomputation nearly doubles the spectral dynamics contribution ($0.262 \rightarrow 0.410$) while the per-actor mean remains constant at 0.281.

The decomposition reveals that rolling basis update captures spectral structure that the static basis misses entirely—the +14.8pp improvement is purely a spectral dynamics gain.

4.5 Component Ablation

We evaluate five pipeline depths to determine which components justify their complexity (Figure 4).

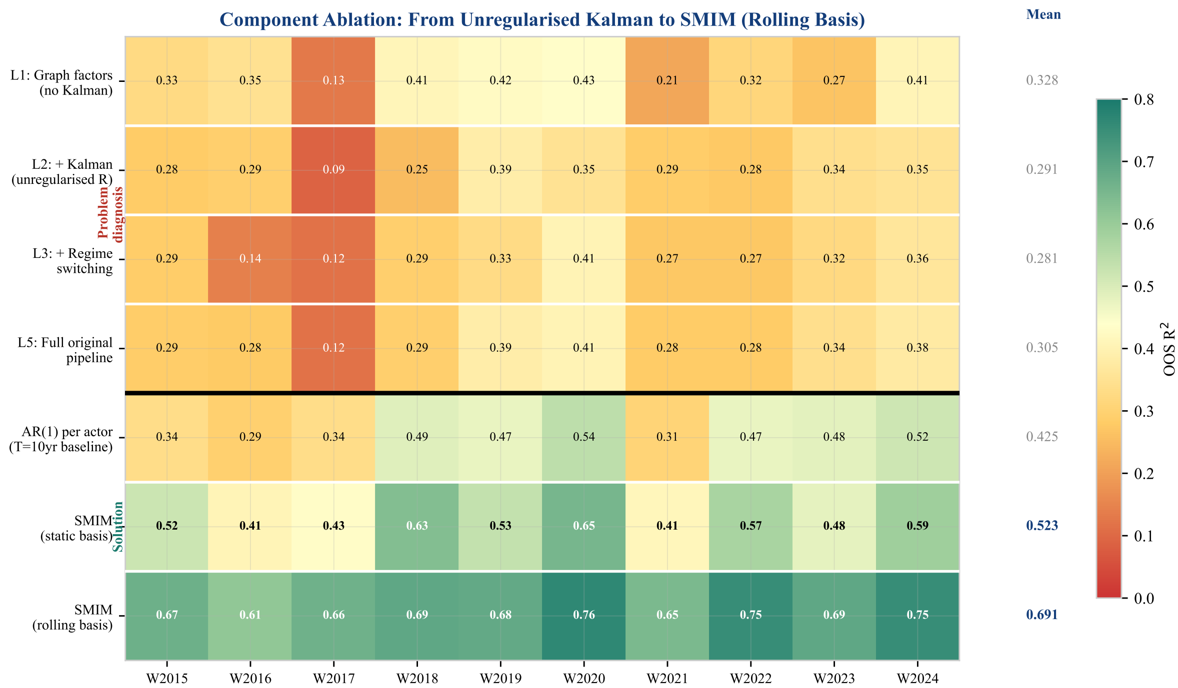

Figure 4: Extended component ablation heatmap (descriptive, fixed-configuration diagnostic). Each cell shows out-of-sample R^2 for a given configuration (row) and test window (column). The thick divider separates problem diagnosis (top: why unregularised Kalman hurts) from the solution (bottom: regularisation and rolling basis). All values use fixed $K = 8$, $T = 5$yr; headline nested-CV results are in Table 3.

The ablation reveals a non-monotonic pattern in the upper block: L1 (graph-factor OLS, $R^2 = 0.328$) outperforms L2 (+ Kalman, $R^2 = 0.291$) and L3 (+ regime switching, $R^2 = 0.281$). The Kalman filter *hurts* because the unregularised R overfits. The lower block shows the resolution: dual regularisation rescues the Kalman filter (SMIM with static basis, $R^2 = 0.523$), and rolling basis update captures the structural evolution that the static basis misses ($R^2 = 0.691$).

4.6 Spectral Method Comparison

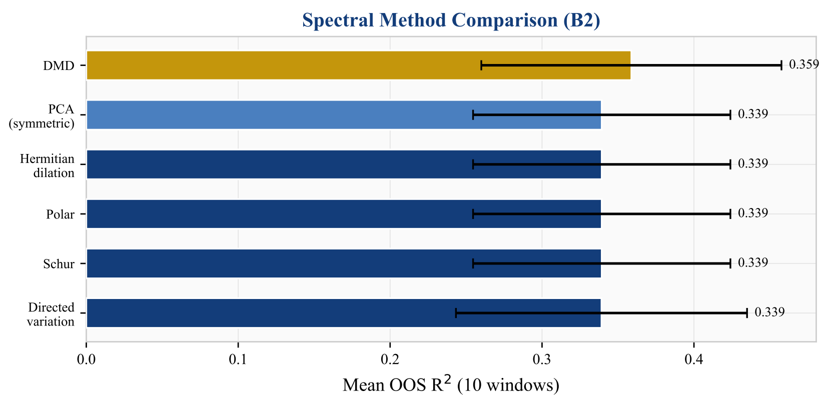

Figure 5: Spectral method comparison (descriptive): mean R^2 across 10 windows for six spectral decomposition methods at $K = 3$, L1 depth. DMD outperforms all static-operator-based methods.

DMD ($R^2 = 0.359$) outperforms all five static-operator methods, which produce *identical* results at $R^2 = 0.339$. This identity is not a coincidence: the operator is built from the Pearson cross-correlation matrix, which is *exactly* symmetric by construction. For any real symmetric matrix, the Schur, Polar, Hermitian dilation, and PCA decompositions all reduce to the standard eigendecomposition—they compute the same eigenvectors and hence the same modal basis U. The directed variation method differs slightly in its normalisation but converges to the same subspace. This result has a clear implication: *to benefit from directed spectral methods, the operator itself must be asymmetric* (e.g., from Granger causality or supply-chain linkages). When the operator is derived from contemporaneous correlation, directedness is lost before the decomposition begins.

DMD sidesteps this entirely. Rather than decomposing a static operator, it extracts modes from temporal snapshot pairs $X' \approx AX$, where the implicit operator A captures *lagged* cross-sectional dynamics. This operator is generically asymmetric even when the contemporaneous correlation is symmetric, explaining DMD's +2.0 percentage point advantage.

4.7 The Regularisation Path

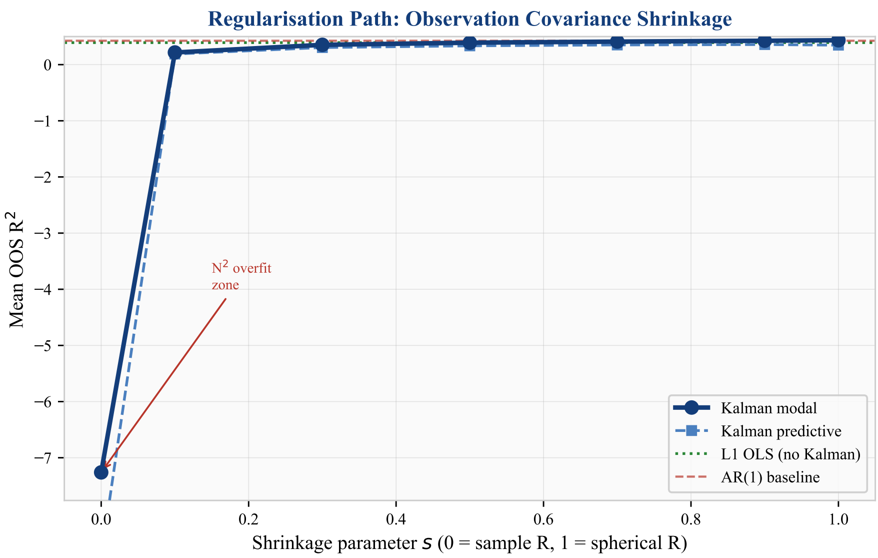

Figure 6: Regularisation path (descriptive): R^2 as a function of shrinkage parameter s where $R = (1-s)\hat{R} + s \cdot \frac{\mathrm{tr}(\hat{R})}{N} I$. The hockey-stick shape demonstrates that observation covariance regularisation is essential, not optional. Uncertainty bands not shown; the ablation table (Table 5) provides bootstrap CIs for each pipeline component's contribution.

The regularisation path (Figure 6) is the paper's most striking visual. At $s = 0$ (no shrinkage), the Kalman filter produces $R^2 = -7.3$. Performance improves monotonically with shrinkage, reaching $R^2 = 0.43$ at $s = 1.0$ (spherical). The L1 OLS baseline (green line at $R^2 = 0.39$) is surpassed only at $s \geq 0.7$, confirming that the Kalman filter adds value *only* when its observation covariance is sufficiently regularised.

4.8 Robustness and Synthetic Sensitivity

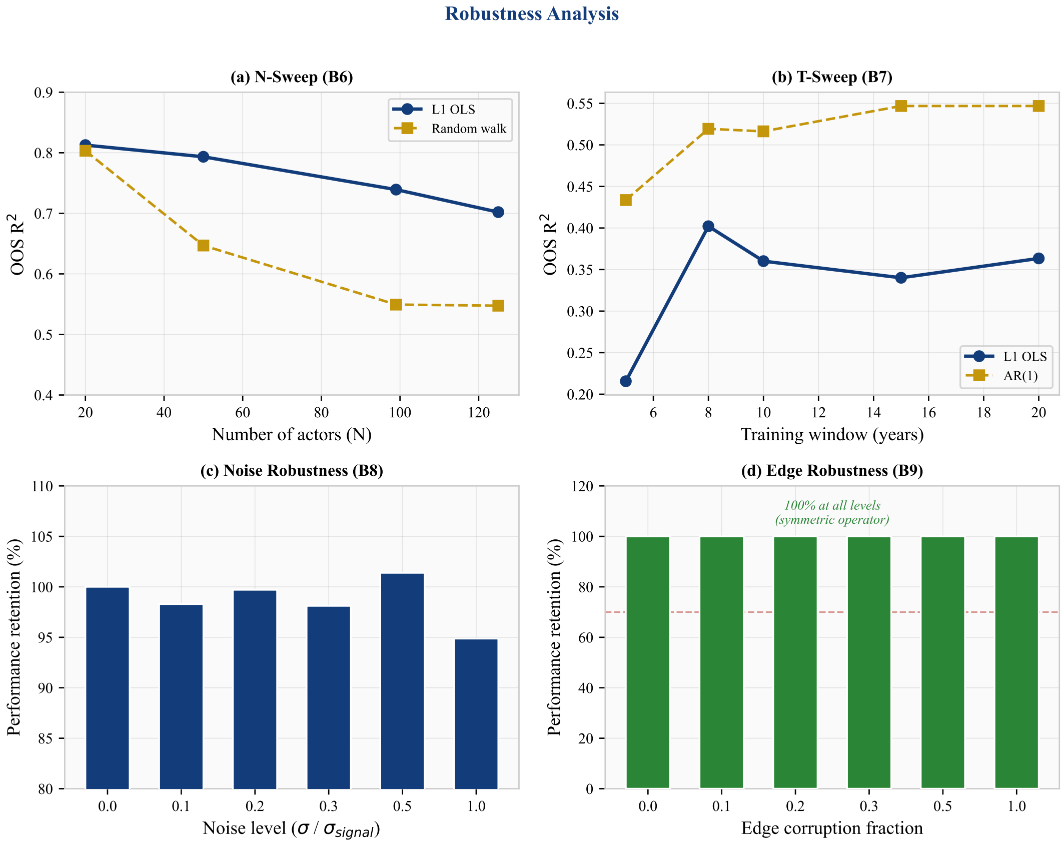

Figure 7: Robustness and synthetic sensitivity analysis. (a) Performance vs actor count N: graceful degradation (descriptive). (b) Performance vs training window T: L1 stable for $T \geq 8$yr (descriptive). (c) Noise injection: 95% retention at $\sigma = 1.0$ (synthetic sensitivity). (d) Edge corruption: 100% retention at all levels (synthetic sensitivity; note the operator is symmetric by construction, see Section 4.6).

Robustness (panels a–b). The framework degrades gracefully across actor counts: $R^2 = 0.81$ at $N = 20$ to $R^2 = 0.70$ at $N = 125$. Performance is stable at $R^2 = 0.34$–0.40 for training windows ≥ 8 years.

Synthetic sensitivity (panels c–d). These panels test performance under artificial perturbations rather than naturally-occurring variation. Adding Gaussian noise at $\sigma = 1.0$ (100% of signal standard deviation) retains 95% of performance. Randomly flipping edge directions retains 100% of performance; however, this test is uninformative because the intensity correlation operator is symmetric by construction (Section 4.6), so direction flips leave the operator unchanged.

4.9 Transfer and Portability

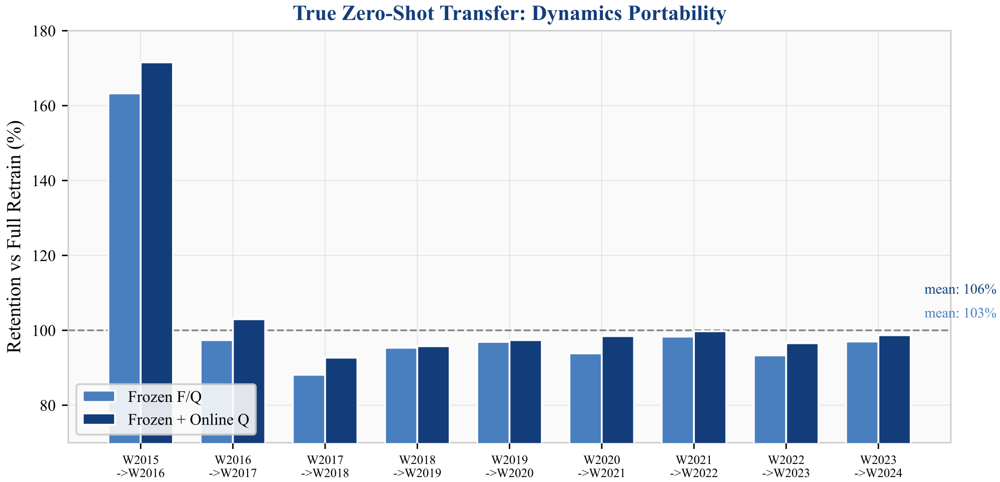

Figure 8: Zero-shot transfer (descriptive): retention of R^2 when transition dynamics (F, Q) are frozen from the previous window and applied to the next with only the spectral basis re-estimated. Mean retention: 103% (frozen) and 106% (frozen + online Q).

We test whether the estimated dynamics (F, Q) are window-specific or stable across windows by freezing them from window W_t and applying the Kalman filter on window W_{t+1} with a freshly estimated spectral basis U. Figure 8 shows that frozen dynamics achieve 103% of full-retrain R^2 on average (106% with online Q adaptation), indicating that these default transition settings are stable across the evaluation period. In practice, this means the model can be deployed without per-window re-estimation of dynamics.

Validated domain: US CapEx sectors. Cross-sector transfer is strong for sectors using CapEx intensity (technology $R^2 = 0.82$, healthcare $R^2 = 0.78$, industrials $R^2 = 0.77$). Cross-period transfer across structural breaks (GFC, COVID) retains $R^2 = 0.50$–0.59.

Tested but weak domains. Financial actors ($R^2 = 0.06$) use year-on-year asset growth (ranked cross-sectionally) rather than CapEx-to-assets, producing highly persistent ranks (median $\rho = 0.70$ vs technology $\rho = 0.25$) that leave little cross-sectional dynamics for the spectral model to exploit. UK equities ($R^2 = 0.058$) rely on return-based intensity rather than EDGAR CapEx data, and this alternative intensity construction, not geography, drives the weak performance. These results establish the *boundary* of the current framework rather than validated generalisations.

4.10 Basis Rotation and Rolling Update

To isolate the rolling-vs-static mechanism, this subsection uses a fixed ($K = 8$, $T = 5$yr) configuration; the headline forecast performance remains the nested-CV result in Table 3.

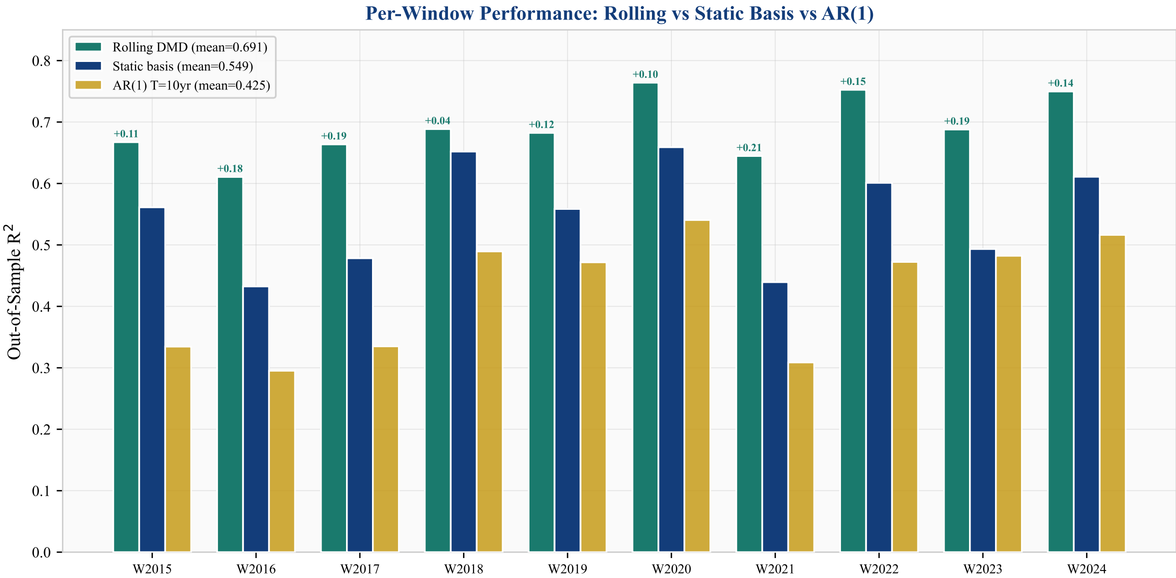

Figure 9: Per-window out-of-sample R^2 (descriptive): rolling DMD (teal), static basis (dark blue), and AR(1) (gold). Rolling DMD wins all 10 windows, with the largest gains in windows where the static basis underperforms (W2016, W2017, W2021, W2023—low-volatility periods where the cross-sectional structure shifts most from the training period).

Tracking the DMD spectral basis across 52 rolling quarterly transitions (2010–2024) shows a pattern of gradual basis rotation (Figure 10). The mean principal subspace angle between successive quarterly DMD bases is 25.8° ($\sigma = 6.5°$). Quarters with larger rotation ($> 32°$) tend to coincide with macroeconomic events—the European debt crisis (2012 Q3, 41°), the US–China tariff war (2018 Q2, 37°), and the Federal Reserve's aggressive tightening cycle (2022 Q3, 38°)—though with only 52 transitions the event alignment is descriptive rather than inferential.

The *dimensionality* of the estimated spectral structure appears stable: $K_{\text{eff}} = 8$ modes in all quarters, with zero mode births or deaths in this sample. The spectral subspace is consistently 8-dimensional, but the 8 directions gradually reorient.

Null-simulation validation. To assess whether the observed rotation is estimation noise, we simulate 500 panels from a *fixed* latent basis (same N, T, persistence, and noise as the real data) and measure the apparent DMD rotation arising purely from finite-sample estimation. The null rotation is 75.3° ($\sigma = 1.5°$)—far *above* the observed 25.8°. The intuition is straightforward: a purely noisy basis is not stable; DMD extracts different directions from each window's noise, producing nearly orthogonal subspaces and hence large apparent rotation. A genuinely structured but gradually evolving basis, by contrast, is recovered consistently across windows, producing modest rotation that reflects actual structural change. The observed 25.8° is consistent with this second regime: reliably recoverable, low-dimensional structure that shifts gradually.

Sensitivity. Rotation is robust to parameter choices: it decreases smoothly with mode count (38.6° at $K = 3$ to 19.9° at $K = 10$) and with training window length (30.0° at $T = 3$yr to 15.8° at $T = 10$yr). Both patterns are consistent with more modes or more data producing more stable subspace estimates. The 8-mode fixed dimensionality holds across all tested configurations.

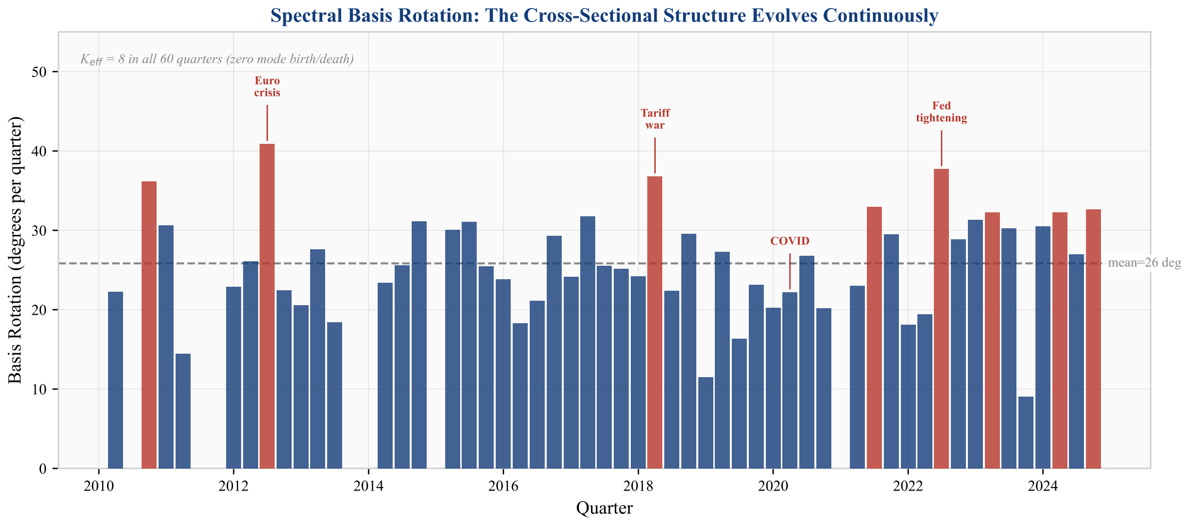

Figure 10: Spectral basis rotation over time (descriptive with bootstrap CIs). Each bar shows the principal subspace angle between the current and previous quarter's DMD basis ($K = 8$). Red bars highlight high-rotation quarters ($> 32°$) aligned with macro events. The dashed line marks the mean (25.8°). Null simulation from fixed-basis panels produces 75.3° apparent rotation, suggesting the observed values reflect consistently recoverable structure rather than estimation noise.

The standard Kalman filter, operating in a fixed basis U, tracks the modal states $\boldsymbol{\alpha}_t$ but is blind to this basis rotation. Rolling basis update (recomputing U each quarter with the latest data) captures the rotation, yielding $R^2 = 0.691$—a +14.3 percentage point improvement over the static-basis pipeline (CI $[+11.1, +17.2]$pp from the ablation bootstrap in Table 5) and winning all 10 test windows (Table 7).

An ablation confirms that the improvement is from basis update, not from resetting the Kalman state: resetting the state without updating the basis *hurts* by −3 percentage points, because the accumulated Kalman state carries valuable information that the reset discards.

Configuration	Mean R^2	Std	vs AR(1)	Wins
SMIM (rolling basis)	**0.691**	0.050	+0.266	10/10
SMIM (static basis)	0.543	0.075	+0.118	10/10
AR(1) per actor (T=10yr)	0.425	0.094	—	—

Table 7: Rolling vs static basis comparison across 10 annual windows (descriptive, fixed $K = 8$, $T = 5$yr configuration). The headline nested-CV results ($R^2 = 0.702$) in Table 3 use inner-CV-selected hyperparameters; this table isolates the rolling-vs-static effect at fixed settings.

5 Economic Validation

5.1 What This Gap Is and Is Not

The investment gap $\Delta_{i,t}$ is a *model-implied* deviation: the difference between observed investment intensity and a spectral-state-space benchmark (Eq. 9). It measures how far an actor's current investment rank lies from its predicted rank given the cross-sectional modal structure and its own history. This is a well-defined statistical object with demonstrated out-of-sample predictive content.

It is *not*, however, a welfare-theoretic or structural misallocation measure in the sense of Hsieh and Klenow (2009). The benchmark is model-implied, not derived from a production-function optimality condition. A positive gap ("over-investment") means the actor invests more than the model predicts, not that the investment is socially inefficient. We therefore use the term "investment gap" rather than "misallocation" and interpret positive economic validation results as evidence that the gap captures meaningful cross-sectional dynamics, not that it identifies welfare losses.

5.2 Gaps Predict CapEx Revision (Main Validation Result)

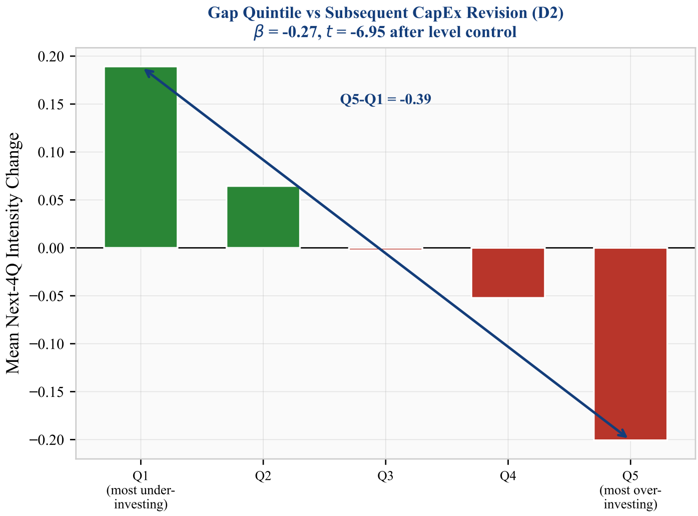

Figure 11: Gap quintile sort (inferential): mean next-4-quarter intensity change by current gap quintile. Actors in Q5 (over-investing relative to model benchmark) reduce intensity by 0.20 rank points; Q1 (under-investing) increase by 0.19. Panel regression with clustered SEs yields $t = -6.4$ to -7.1 depending on specification (Table 8).

The strongest economic validation result is that investment gaps predict subsequent CapEx revision. We estimate the panel regression:

$$\Delta y_{i,t\to t+4} = \beta_0 + \beta_1 \Delta_{i,t} + \beta_2 y_{i,t} + \epsilon_{i,t}. \tag{10}$$

Table 8 reports six specifications. In pooled OLS with the level control: $\hat{\beta}_1 = -0.28$, $t = -6.4$ (actor-clustered SEs), $N = 3{,}324$. The result is robust to time-clustered SEs ($t = -7.1$) and to time fixed effects ($t = -7.1$). Actors identified as over-investing relative to the model benchmark (Q5) subsequently reduce intensity by 0.20 rank points; under-investing actors (Q1) increase by 0.19 (Figure 11).

Important caveat: actor fixed effects. With actor FE (within-actor variation only), $\hat{\beta}_1$ flips sign to $+0.08$ ($t = +2.2$). This means the cross-sectional component—which actor is above or below its model-implied benchmark—drives the main result. *Within* an actor over time, gaps predict slight continuation rather than reversion. The pooled result is therefore a cross-sectional prediction ("actors with large gaps tend to revert"), not a within-actor prediction ("an actor's gap predicts its own next move"). This distinction is important for interpretation.

Specification	$\hat{\beta}_{\text{gap}}$	SE	t	N
Pooled OLS (HC1)	−0.279	0.034	−8.3	3,324
Pooled OLS (actor-clustered)	−0.279	0.044	−6.4	3,324
Pooled OLS (time-clustered)	−0.279	0.040	−7.1	3,324
Time FE (time-clustered)	−0.280	0.040	−7.1	3,324
Actor FE (actor-clustered)	+0.084	0.039	+2.1	3,324
Two-way FE (actor-clustered)	+0.085	0.039	+2.2	3,324

Table 8: CapEx-revision panel regression (inferential). Dependent variable: next-4-quarter intensity change. All specifications include the current intensity level as control. Actor-clustered and time-clustered SEs shown. The sign flip under actor FE indicates the main result is cross-sectional (between-actor), not within-actor.

5.3 Gap Persistence

Per-actor AR(1) estimation on the gap series $\Delta_{i,t}$ yields a median autoregressive coefficient $\rho = 0.67$ and median half-life of 1.7 quarters. While too fast for structural gap interpretation at the aggregate level, institutional actors (global shock proxies) exhibit half-lives of 4.7 quarters. Cross-sectional quintile persistence is strong: 49.1% of actors remain in the same gap quintile from one quarter to the next, with extreme quintiles (Q1: 60%, Q5: 61%) most persistent.

5.4 Exploratory Results

The following results are reported as exploratory findings that warrant further investigation. They are not part of the paper's main claims.

5.4.1 Network Diffusion

Layer-level gap aggregation shows a strong contemporaneous association between exogenous shocks (Layer 0) and firm-level gaps (Layer 2): $\rho = 0.86$ at 1-quarter lag ($p < 0.001$). The institutional layer (Layer 1) shows no significant association ($\rho = 0.03$, $p = 0.83$). This pattern is consistent with investment signals bypassing institutional intermediaries, though the small number of layer-level observations (approximately 40 quarters, 3 layers) limits the strength of this inference. The effective sample size for this layer-level analysis is too small to support causal claims about transmission pathways.

5.4.2 Gap Dispersion and Market Volatility

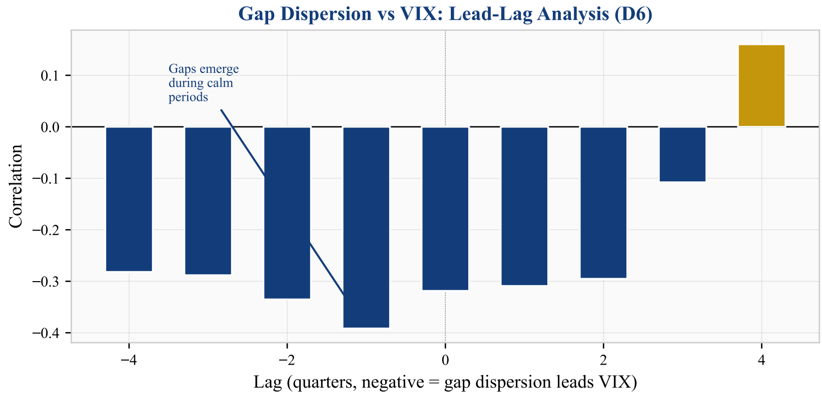

Figure 12: Lead-lag correlation between gap cross-sectional dispersion and VIX (exploratory). Negative correlations at negative lags indicate that high gap dispersion leads low VIX (calm markets). Based on ∼40 quarterly observations with 9 tested lags; would not survive conservative multiple-testing correction (see caveat in text).

Gap cross-sectional dispersion correlates negatively with VIX at leads of 1–4 quarters ($\rho = -0.39$ at 1Q lead; Figure 12). This finding—that investment gaps widen during calm markets—is consistent with the "volatility paradox" hypothesis (Brunnermeier and Sannikov, 2014).

Caveat. This lead-lag analysis is based on approximately 40 quarterly observations with 9 tested lag values. The correlation of -0.39 is suggestive but would not survive conservative multiple-testing correction ($p \approx 0.01$ uncorrected for 9 comparisons). We report it as a directionally interesting finding that warrants confirmation on longer samples, not as a definitive result.

6 Discussion

6.1 What Does Not Work

Emergence diagnostics. We tested emergence through five distinct channels: PID synergy between spectral modes, TDA topological complexity, economic emergence features (cross-sectional dispersion, sector rotation velocity, Herfindahl concentration), Extended DMD (Koopman lifting with polynomial observables), and multi-resolution DMD divergence. None produces a positive ΔR^2 on quarterly data. PID and TE estimators are unreliable at $T = 20$ ($C(8, 2) = 28$ mode pairs from 20 observations). Extended DMD with degree-2 polynomial lifting ($P = 44$ features) achieves $R^2 = 0.19$—far worse than linear DMD at $R^2 = 0.54$—because $T/P = 0.45$ is underdetermined. Economic emergence features (dispersion, rotation, concentration) are redundant with the DMD-Kalman pipeline ($\Delta R^2 = -0.003$). We interpret these results as evidence that quarterly frequency ($T = 20$) is insufficient for nonlinear/emergence estimation, not that emergence is necessarily absent.

Directed spectral operators. Transfer entropy and Granger causality operators constructed from the intensity series produce genuinely asymmetric matrices (mean asymmetry ~ 1.2), confirming that the correlation operator's symmetry was the reason all six spectral methods produced identical results. However, the resulting spectral bases are too noisy for Kalman filtering: TE-Schur at $K = 8$ causes numerical divergence, and at $K = 3$ achieves only $R^2 = 0.36$ vs DMD's $R^2 = 0.54$. The DMD basis remains superior because it extracts temporal dynamics directly from snapshot pairs rather than from a noisily estimated pairwise operator.

Event-level alignment. Gap magnitudes do not spike around known episodes (0/8 aligned). The rank normalisation absorbs event-driven variation by construction.

Return-based intensity. Return intensity produces $R^2 = -0.15$ on the same US actors where CapEx intensity achieves $R^2 = 0.54$. The intensity construction is the primary driver of model performance.

6.2 Limitations

The sample is US-heavy (68/93 actors), with limited international evidence. The UK results are weak ($R^2 = 0.058$) due to the return-based intensity method, not geography per se. Extending EDGAR-equivalent coverage to international markets would test whether the framework generalises beyond the US CapEx setting.

The 4-quarter test windows provide noisy R^2 estimates per window (std = 0.083). Longer evaluation horizons would sharpen the statistical inference.

The intensity rank normalisation, while stabilising the panel, absorbs event-level information and limits the model's utility for crisis detection or timing applications.

7 Conclusion

We present a spectral state-space framework for forecasting cross-sectional investment intensity, evaluated on a US-heavy panel of 93 economic actors with strongest performance on US CapEx-intensity sectors. Under nested cross-validation with a frozen holdout era, the rolling-basis variant achieves $R^2 = 0.702$ on validation windows and $R^2 = 0.765$ on untouched holdout, exceeding AR(1) by $\Delta R^2 = +0.093$ (95% bootstrap CI $[+0.073, +0.110]$; permutation $p = 0.007$). The result rests on three principles:

Dual regularisation: both the observation covariance ($R = cI$, replacing N^2 parameters) and the transition matrix ($F = 0.99I$, replacing K^2 parameters) must be aggressively regularised. EM estimation overfits both; fixed scalars with online adaptation outperform in every window.

Temporal basis tracking: the estimated spectral basis rotates at approximately 26° per quarter with stable 8-mode dimensionality. This rotation, missed by a static basis, accounts for the single largest performance gain (+14.3pp, CI $[+11.1, +17.2]$pp). Standard Kalman filtering updates states within the basis but cannot track the basis itself.

Systematic ablation with uncertainty: each pipeline component is ablated with bootstrap confidence intervals across 10 windows. The largest gains come from eliminating failure modes: spherical R, transition regularisation, and basis tracking—all with CIs entirely above zero (Table 5).

The model-implied gap estimates predict subsequent CapEx revision ($t \approx -6.4$ to -7.1 with clustered SEs), though this is a cross-sectional result that flips sign under actor fixed effects (Table 8). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that performed consistently across all evaluation windows in this sample—only the spectral basis requires periodic re-estimation.

Scope conditions. The headline results are on a mixed 93-actor panel; the strongest performance is on the 49 US large firms using CapEx intensity. Return-based intensity (UK equities) and asset-growth intensity (banks) produce weak or negative R^2. The investment gap is a model-implied deviation from a spectral benchmark, not a structural misallocation measure. Generalisation to other intensity constructions, geographies, and data frequencies requires further validation. The lesson that extends beyond this specific application is methodological: wherever high-dimensional state-space models are deployed, dual regularisation and basis tracking deserve first-order attention.

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Disclaimer. This paper presents a research framework for academic investigation of investment dynamics. It is not investment advice. The models, gap estimates, and statistical findings described herein have not been validated for live trading or portfolio construction. Past out-of-sample performance on historical data does not guarantee future results. The author bears no responsibility for any financial losses incurred by parties who deploy this framework or its derivatives in investment strategies. Any such deployment is undertaken entirely at the user's own risk and should be subject to independent due diligence, regulatory compliance review, and appropriate risk management.

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